MIL-I-82532(OS)
18 April 1968
Superseding
NOTS XS 484 Rev. E
16 November 1967

MILITARY SPECIFICATION
FOR
IGNITER, ROCKET MOTOR
MARK 251 MOD 0

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements of one type of Igniter, Rocket Motor Mark 251 Mod 0. (See 6.1 for intended use and description.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein except as noted.

SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-C-843	Cushioning Material, Cellulosic

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3955	Cans, Fiber, Spirally Wound
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

FSC -1340

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MIL-L-10547	Liners. Case, and Sheet, overwrap; Water-Vaporproof or Waterproof, Flexible

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels

MILITARY

MIL-STD-105D	Sampling Procedures and Tables for Inspection by Attributes
IWIL-STD-129	Marking for Shipment and Storage
MIL-STD- 167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 497905	Igniter, Rocket Motor, Mark 251 Mod 0 and all drawings listed thereon

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications

INTERSTATE COMMERCE COMMISSION

49 CFR Part 71-90	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations and are available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite latest edition and revision thereto.)

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3. REQUIREMENTS

3.1 General requirements. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the igniter shall be in accordance with the requirements specified herein and with the applicable documents listed in LD 497905.

3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the igniter shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.3 Performance and product characteristics. The igniter shall meet the following performance and product characteristics.

3.3.1 Nonfunctioning

3.3. 1.1 Electrical continuity and resistance. The electrical circuits of each of the squibs shall be continuous, with resistances not less than 0.15 ohm nor more than 0.22 ohm.

3.3.1.2 Insulation resistance. The resistance between the shorted lead wires and the igniter case shall be not less than 20 megohms when a potential of 500 plus or minus 50 volts DC is applied prior to being subjected to the environmental conditions of humidity test (3.4.4), fungus test (3.4.5), and salt spray test (3.4.6).

3.3.1.3 Airtightness. The leak rate of the igniter shall be not greater than 0.00001 cubic centimeter per second when subjected to a pressure differential of 2.00 plus or minus 0.25 pounds per square inch (psi).

3.3.2 Functioning

3.3.2.1 Case-opening time. The case-opening time as defined in 6.3.1, shall be not less than 3 milliseconds (ins), nor more than 75 ms from the time of initial application of an ignition current to each squib circuit equivalent to a DC voltage of 2.0 plus or minus 0.1 volts across a 0.22 plus or minus 0.01 ohm resistor.

3.3.2.2 Flame duration. The total flame duration of the initiated igniter, as defined in 6.3.2, shall be not less than 50 ms nor more than 400 ms.

3.3.2.3 Minimum peak intensity The minimum peak intensity, as defined in 6.3.3, shall be not less than 500, 000 candlepower.

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3.4 Environmental requirements. Unless otherwise specified, the igniter shall be capable of conforming to the performance and product characteristics of this specification after being exposed to the environments and all conditions specified herein.

3.4.1 Vibration. The igniter shall withstand vibration as specified for Type I equipment in MIL-STD-167 up to 25 cps (cycles per second).

3.4.2 Shock, standby condition. The igniter shall withstand shocks of 40g (g = units of gravitational acceleration) peak magnitude. The shocks shall have a pulse shape approximating a one-half sine wave having a duration of 5 ms.

3.4.3 Temperature, storage condition. The igniter shall withstand temperatures from minus 65 to plus 160 degrees F (Fahrenheit).

3.4.4 Humidity. The igniter shall withstand relative humidity up to 95 percent as specified in MIL-E-5272, Procedure I, and thereafter shall meet the performance and product characteristics of this specification, except the requirement for insulation resistance as specified in 3.3.1.2 and 4.6.1.1.2.

3.4.5 Fungus. The igniter shall withstand fungus conditions as specified in MIL-E-5272, Procedure I, and thereafter shall meet the functional characteristics of 3.3.2.

3.4.6 Salt spray. The igniter shall withstand salt spray conditions as specified in MIL-E-5272, Procedure I, and thereafter shall meet the functional characteristics of 3.3.2.

3.4.7 Temperature, operating condition. The igniter shall meet the functional characteristics of 3.3.2 while at temperatures from 35 to 120 degrees F.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the

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supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the igniter shall be classified as follows:

a. Preproduction and periodic production inspection (See 4.5.)

b. Quality conformance inspection (See 4.6.)

4.3 Sampling. (See 6.2.1.)

4.3.1 Lot size. The term "lot" shall mean "inspection lot"; that is, a collection of product submitted by a supplier for inspection. Unless otherwise specified in the contract or order, the number of product in each lot shall be three hundred (300).

4.3.2 Preproduction sample. A preproduction sample of twenty (20) igniters, manufactured in accordance with 3.2, shall be subjected to the preproduction tests detailed in 4.5. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the igniters by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.3 Periodic production sample. From each succeeding lot of three hundred (300) or less, a random sample of ten (10) igniters shall be selected. These sample units shall be subjected to the tests of 4.5. The igniters shall be considered satisfactory if they meet all specification and pertinent drawing requirements and withstand the tests specified in 4.5. (See Table I.) When a defective unit is found in any lot, acceptance of all units in the lot shall be withheld and the contractor shall:

a Determine the cause of the defect.

b. Take the necessary action, with the concurrence of the procuring activity, to correct or eliminate the defect from all units in the lot and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

4.3.4 Quality conformance inspection sampling. Samples shall be selected as specified in 4.6 and shall be used to determine the acceptability of the igniters inspected as specified in 4.6.

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Table I. Preproduction and Periodic Test Sequence

Test	Paragraph Number	Unit Quantities Preproduction	Periodic
Quality Conformance Inspection:	4 6		
Continuity and resistance	4 6 1 1 1	20	10
Insulation resistance	4.6 1 1 2	20	10
Airtightness	4 6 1.1 3	20	10
Vibration	4 5 1 1	20	10
Quality Conformance Inspection:	4 6		
Continuity and resistance	4.6 1 1 1	20	10
Insulation resistance	4.6.1.1.2	20	10
Shock, Standby Condition	4.5.1.2	20	10
Quality Conformance Inspection:	4.6		
Continuity and resistance	4.6.1.1 1	20	10
Insulation resistance	4.6.1 1 2	20	10
Airtightness	4.6.1.1.3	20	10
Temperature, Storage Condition	4.5.1.3	20	10
Quality Conformance Inspection:	4.6		
Continuity and resistance	4.6.1.1 1	20	10
Insulation resistance	4 6 1 1 2	20	10
Airtightness	4 6 1 1 3	20	10
Humidity	4.5.1 4	20	10
Quality Conformance Inspection:	4.6		
Continuity and resistance	4 6 1 1 1	20	10
Airtightness	4.6 1 1 3	20	10
Fungus	4 5 1 5	5	-
Quality Conformance Inspection:	4.6		
Functioning	4 6 1 2	5*	
Salt Spray	4 5.1 6	5	2
Quality Conformance Inspection:	4.6		
Functioning	4 6.1 2	5*	2*
Temperature, Operating Condition	4 5 1.7		
Low Extreme	4 5 1 7 1	5	4
Quality Conformance Inspection:	4.6		
Functioning	4.6 1 2	5*	4*
High Extreme	4.5.1 7 2	5	4
Quality Conformance Inspection:	4 6		
Functioning	4 6.1.2	5*	4*

*Samples are expended during functioning.

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4.4 Test equipment and test conditions.

4.4.1 Quality conformance inspection equipment. The following items of test equipment are required to perform the quality conformance inspection tests:

Equipment	Capability
a. Ohmmeter, current limited	Measure 0.00 to 0.50 ohm, accuracy plus or minus 0.01 ohm; maximum current limited to 10 milliamperes,
b. Megohmmeter	Measure 0.5 to 50 megohms with 500 plus or minus 50 volts DC potential; full scale accuracy of 10 percent.
c. Leak detector	Detect gas leakage equivalent to one ten-thousandth of a cubic centimeter per second.
d. Light meter	Vacuum or gas type photocell with amplifier having joint frequency response of not less than 200 cps; measure light intensities of 2 x 10^6 candlepower with accuracy of plus or minus 10 percent.
e. Electronic counter	Measure time interval between two events in units as short as 500 microseconds.
f. Oscillograph	Record output from light meter specified in "d" above; have a paper speed of 40 inches per second or faster; provide a timing signal, in milliseconds, on paper.
g. Power supplies	Two identical and independent 60-cps power supplies capable of supplying sufficient current to cause a voltage drop of 2.0 plus or minus 0.2 volts across a load of 0.22 plus or minus 0.01 ohm.

4.4.2 Preproduction and periodic production test equipment. The following items of test equipment are required to perform the preproduction and periodic production tests. These items are in addition to selected items of quality conformance inspection equipment and pieces of equipment recommended by applicable specifications or standards referenced within test procedure paragraphs.

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Equipment	Capability
a. Shock equipment	Apply shocks of 40 plus or minus 4g; a one-half sine wave pulse shape of 5 plus 3 minus 0 ms duration; maintain direction along specified axes,
b. Temperature chamber	Maintain temperature at: Minus 65 plus 5 minus 0 degrees F Plus 35 plus 5 minus 0 degrees F Plus 120 plus 0 minus 5 degrees F Plus 160 plus 0 minus 5 degrees F

4.4.3 Test conditions. Unless otherwise specified, the igniter shall be subjected to the preproduction and periodic production tests and quality conformance inspection tests under the following conditions:

a. Temperature	Room ambient (65 to 95 degrees F)
b. Altitude	Normal ground atmospheric pressure
c. Humidity (Relative)	Room ambient up to 95 percent

4.5 Preproduction and periodic production inspections. The preproduction and periodic production samples shall be subjected to the tests specified in 4.5.1 after satisfactorily passing the quality conformance inspections of 4.6.1.1, 4.6.1.2, and 4.6.1.3.

4.5.1 Test procedure. Subject the igniter to the test sequence listed in Table I. Unless otherwise specified in the individual test paragraph, the quality conformance inspection as specified in Table I shall be performed subsequent to environmental tests under test conditions specified in 4.4.3. Failure of the igniter to pass any one of these tests shall constitute a defective unit.

4.5.1.1 Vibration. The igniter shall be subjected to the vibration tests for Type I equipment as specified in MIL-STD-167 up to 25 cps in each of any three mutually perpendicular axes.

4.5.1.2 Shock, standby condition. The igniter shall be subjected to shocks of 40 plus or minus 4g peak magnitude, with a pulse shape approximating a one-half sine wave and a duration of 5 plus 3 minus 0 ms Shocks shall be applied three times in each direction along each of any three mutually perpendicular axes.

4.5.1.3 Temperature, storage condition. The igniter shall be subjected to a temperature of minus 65 plus 5 minus 0 degrees F for not less than 48 hours. The temperature shall be raised to plus 160 plus 0 minus 5 degrees F in not less than 2 hours and maintained at this temperature for not less than 48 hours. The

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unit shall then be subjected to room temperature for not less than 4 hours prior to performing subsequent tests.

4.5.1.4 Humidiy. The igniter shall be subjected to Humidity test, Procedure I, as specified in MIL-E-5272. Thereafter the igniter shall pass the quality conformance inspection tests, except the insulation resistance test of 4.6.1.1.2

4.5.1.5 Fungus. Unless otherwise specified in the contractor order, this test shall be performed on preproduction samples only. The igniter shall be subjected to Fungus test, Procedure I, as specified in MIL-E-5272. Thereafter the igniter shall pass the functioning test of 4.6.1.2.

4.5.1.6 Salt spray. The igniter shall be subjected to Salt Spray test, Procedure I, as specified in MIL-E-5272. Thereafter the igniter shall pass the functioning test of 4.6.1.2.

4.5.1.7 Temperature, operating condition.

4.5.1.7.1 Low extreme. The igniter, selected from the sample units as indicated in Table I, shall be subjected to a temperature of 35 plus 5 minus 0 degrees F for not less than 2 hours and while at that temperature shall pass the functioning test of 4.6.1.2.

4.5.1.7.2 High extreme. The igniter, selected from sample units as indicated in Table I, shall be subjected to a temperature of 120 plus 0 minus 5 degrees F for not less than two hours and while at that temperature shall pass the functioning test of 4.6.1.2.

4.6 Quality conformance inspection. Samples selected using the plans and procedures set forth in Table II of this specification shall be subjected to the tests and examination described in 4.6.1.

4.6.1 Test procedure. The igniter shall be subjected to the tests and examinations outlined herein. Unless otherwise specified, inability of the igniter to pass the tests and examinations shall constitute a failure.

4.6.1.1 Nonfunctioning.

WARNING

To prevent personnel injury due to inadvertent activation of the igniter, use only the current-limited ohmmeter specified in 4.4.1a to perform the continuity and resistance test. Current in excess of 10 milliamperes may activate the igniter.

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Table II. Quality Conformance Tests and Examinations and AQL

Test	Sampling Plan	AQL	Test Paragraph
Continuity and resistance	*	*	4.6 1 1 1
Insulation resistance	*	*	4 6 1.1 2
Airtightness	*	*	4.6 1.1.3
Functioning	**	**	4.6 1 2
Visual examination	*	*	4 6.1.3
Preservation, packaging, packing, and marking	*	*	4.7

*One-hundred percent inspection; defect is cause for rejection of individual unit.
** MIL-STD-105D, Special Inspection Level S4; zero defects accepted.

4.6.1.1.1 Continuity and resistance. Using the ohmmeter specified in 4.4.la, measure the resistance of each blue-white circuit of the exposed igniter cable leads. The electrical circuits shall be continuous and the resistances shall be not less than 0.15 ohm nor greater than 0.22 ohm.

4.6.1.1.2 Insulation resistance. Using the megohmmeter specified in 4.4.1b, measure the resistance between the exposed igniter cable leads, shorted together, and the igniter case. The applied potential shall be 500 plus or minus 50 volts DC, applied for not less than 5 seconds. The resistance between the shorted leads and the case shall be not less than 20 megohms. This test shall be performed prior to humidity test (4.5.1.4), fungus test (4.5.1.5), and salt spray test (4.5.1.6).

4.6.1.1.3 Airtightness. Immediately prior to assembly, the igniter case shall be thoroughly purged with an inert indicator gas. Subsequent assembly operations shall be conducted so as to retain a substantially pure atmosphere of inert gas inside the igniter. Within 20 minutes after sealing, the igniter shall be placed in a partial vacuum equivalent to 25.84 plus or minus 0.50 inches of mercury, absolute, and subjected to leak examination using the leak detector specified in 4.4.1c. The leak rate of the igniter shall be not greater than 0.0001 cubic centimeter per second.

4.6.1.2 Functioning. Subject the igniter to functioning tests using the test equipment specified in 4.4.1d through 4.4.lg. The light meter shall be positioned with respect to the igniter as shown in Figure 1.

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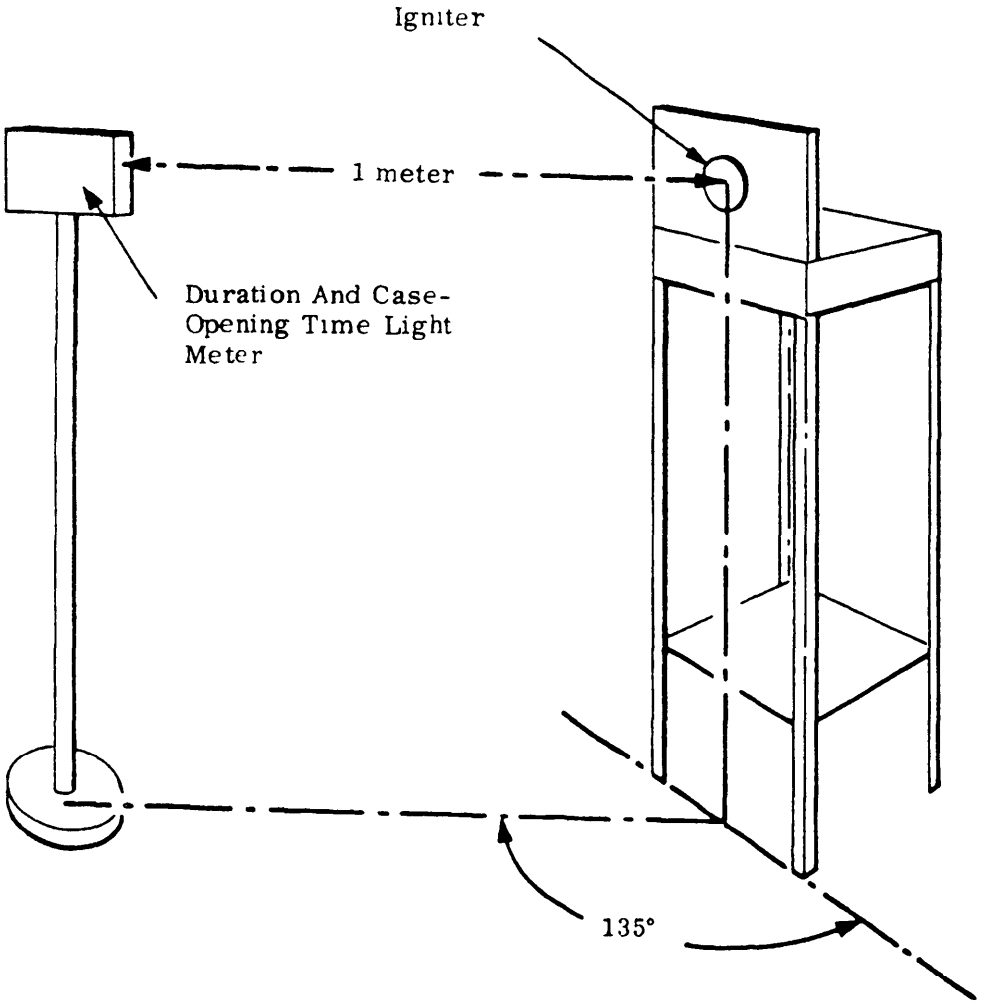


Figure 1. Test Setup for Functioning Tests

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4.6.1.2.1 Test procedure. Simultaneously apply an ignition current equivalent to 2.0 plus or minus 0.1 volts DC across a 0.22 plus or minus 0.01 ohm resistor to each igniter squib circuit. Limits for case-opening time, total flame duration, and peak intensity shall be as specified in Table III.

Table III. Functioning Requirements Acceptance Criteria

Function	Limits Lower	Upper
Case-opening time (ins)	3	75
Total flame duration (ins)	50	400
Peak intensity (candlepower)	500, 000 (minimum)	

4.6.1.3 Visual Examination. Visual examination shall be made on each igniter submitted for delivery to the procuring agency. The igniter shall be examined for conformance to the requirements of good workmanship in 3.5. Nonconformance is cause for rejection of individual unit.

4.7 Preservation, packaging, packing, and marking. The igniter shall be examined for conformance to the prescribed preservation, packaging, packing, and marking requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified. (See 6.2b and 6.4.)

5.1.1 Level A.

5.1.1.1 Cleaning. The igniter shall be cleaned in accordance with Process C-1, MIL-P-116.

5 1.1 2 Drying. The igniter shall be dried in accordance with Procedure D-1 or D-4, MIL-P-116.

5.1.1.3 Unit packaging. One igniter, cleaned and dried in accordance with 5.1.1.1 and 5.1.1.2 shall be placed in a close fitting container conforming to MIL-C-3955. The igniter shall be cushioned with material conforming to PPP-C-843, Type III. Containers shall conform to 49 CFR 73.92.

5.1.2 Level C. The igniter shall be packaged in accordance with the manufacturer's commercial practice. Type of containers shall conform to 49 CFR 73.92.

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5.2 Packing. Packing shall be Level A or C as specified (See 6.2c and 6.4.)

5.2.1 Level A. Unless otherwise specified in the contract, order, or requisition, twelve unit packages, each containing one igniter packaged in accordance with 5.1.1, shall be packed in exterior-type shipping container, PPP-B-621, of uniform size and shape, with the minimum cube and tare consistent with the protection required. The gross weight of each pack shall be limited to approximately 100 pounds Containers shall be provided with a case liner, MIL-L-10547, and shall be closed and strapped in accordance with PPP-B-621. Containers shall conform to 49 CFR 73.92.

5.2.1.1 Cushioning. Each package shall be cushioned with material conforming to PPP-C-843, Type III.

5.2.2 Level C. The igniter, packaged in accordance with 5.1.2, shall be packed or otherwise prepared for shipment to insure carrier acceptance and to insure safe delivery to destination at the lowest applicable rate. Shipping containers shall conform to 49 CFR 73.92, or other carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, and 49 CFR 73.92 (C) and (d).

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The igniter is intended to be used in 11.65 Rocket Motor Mk 1 Mod 0 or Mk 37 Mod 0.

6.1.2 Description. The igniter is an electrically initiated device designed to produce a flame sufficient to ignite the propellent grain of a rocket motor. The igniter consists of a steel case with a pressure tap hole for test purposes in the forward end and a cavity filled with boron potassium nitrate pellets in the aft end. The pellets are retained and protected by a concentrically corrugated metal blowout disc soldered in place. Two electrically initiated squibs projecting into the cavity provide a means of igniting the pellets.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and data of this specification

b. Required level of preservation and packaging

c. Required level of picking

d. Any exceptions to this specification

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6.2.1 Additional samples. Additional samples required because of failure of any sample(s) to pass the prescribed inspections shall be provided and inspected at the expense of the supplier.

6.3 Definitions. For the purpose of this specification the following definitions have been established.

6.3.1 Case-opening time. Case-opening time is defined as the time elapsed between the application of an electrical ignition current and the generation of an ignition flame intensity of 5, 000 candlepower

6.3.2 Flame duration. Flame duration is defined as the interval of time the flame intensity remains above 200, 000 candlepower.

6.3.3 Peak intensity. Peak intensity is defined as the maximum flame intensity produced by the igniter after ignition and case-opening.

6.3.4 Unit condition.

6.3.4.1 Operating. Operating is the condition of the igniter during missile operation.

6.3.4.2 Standby. Standby is the condition of the igniter when the igniter is aboard ship and assembled with the end item.

6.4 Preparation for delivery criteria. Criteria for the use of the proper level of preservation, packaging, and packing shall be as specified in FED-STD-102.

6.5 General safety precautions. The loading, assembly, and handling of explosives, ammunition, ammunition components, fuzes, detonators, primers, tracers, boosters, pyrotechnics, etc., and subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions that a contractor or manufacturer (loader) of the above items should observe in order to fulfill his responsibility for safety At Naval Ordnance Systems Command and other Government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail provisions of the specification.

6.5.1 All loading operations should be conducted in a neat and orderly manner.

6.5.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded

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handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, and crimping.

6.5.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts that affect functioning, should, insofar as practicable, avoid using bare fingers or improper equipment to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intra-plant distances when outside of loading rooms. For Navy explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives shall apply.

6.5.6 Proper care should be exercised at all times to protect personnel from accidents, fires, or explosions, and to limit damage to equipment and loading areas. In this connection, the following precautionary measures should be observed.

6.5.6.1 Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

6.5.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.6.3 Keep explosives and explosive parts in approved, covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.5.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generations should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items that are usually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy-type electric primers, detonators, and squibs. The latter types of items

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should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare, noninsulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short-circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding of these items from induced electric currents generated by sources such as lightning, static radiations from communications apparatus, radar, or high frequency heating apparatus. Where necessary for safety, humidity (see 6.5.4) of work rooms should be increased as required to lessen electrostatic effects without excessive moisture absorption.

6.5.6.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, and test instruments, by suitable insulation, grounding, separation, or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or by completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electronic soldering iron through a grounded contact. All electric-type primers, detonators, or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable removable short circuiting clips, or other devices should be employed during manufacturing operations involving electric primers, detonators, or squibs.

6.5.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning, and associated industrial hazards.

6.5.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper MATCH and NO SMOKING rules where necessary.

6.5.6.8 Enforce good housekeeping and maintain effective policing, inspection, and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping and equipment or the items loaded. Clean up any spilled material immediately.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-82532(OS) IGNITER, ROCKET MOTOR MARK 251 MOD 0

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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DEPARTMENT OF THE NAVY
Naval Ordnance Systems Command
Washington, D.C. 20360

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DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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